# Simple Harmonic Motion

- Restoring force proportional to displacement
- Hooke’s law: F = -kx for an ideal spring
- Angular frequency of a mass-spring system: omega = sqrt(k/m)
- Period of oscillation T = 2 pi sqrt(m/k)

Definition: amplitude is the maximum displacement from equilibrium.

# Energy in Simple Harmonic Motion

- Kinetic and potential energy exchange during each oscillation
- Total mechanical energy E = 1/2 k A^2 remains constant
Derivation: total energy from Hooke’s law and energy conservation
Example: simple pendulum period for small angles T = 2 pi sqrt(L/g)

# Damped Oscillations

- Damping force proportional to velocity reduces amplitude
- Underdamped, critically damped and overdamped motion

Example: calibrating a quartz crystal oscillator in a wristwatch circuit

Example: seismometer magnetic braking with eddy current coils